

BLOCK 7

TRADE, GROWTH AND DEVELOPMENT

Unit 16	Debate on the Trade and Growth Nexus
Unit 17	Trade and Environment

BLOCK INTRODUCTION

This block is on **Trade, Growth and Development**. It has two units, i.e., Unit 16 - **Trade and Economic Growth Nexus** and Unit 17 – **Trade and Environment**.

Unit 16 provides us an understanding of the link between the economic growth and trade. In order to do so, the unit starts with the discussion on the importance of economic growth with reference to the theoretical underpinnings in terms of classical and endogenous models, trade and Solow model, trade and productivity growth. The unit moves in the direction of explaining various trade policy regimes vis-à-vis growth in developing countries wherein the import substitution and outward oriented development strategies are discussed. At the end of the unit, India's experience with the inward-oriented and outward-oriented regime is presented.

Unit 17 takes the theme ahead by providing us the link between trade and environment wherein the role of externalities has been outlined. The unit also discusses the link of trade with climate change and the policies and practices in that regards. The relevance of WTO in safeguarding the environment has been highlighted and, in the end, the multilateral environmental agreements in relation to trade are discussed.

THE PEOPLE'S
UNIVERSITY

UNIT 16 DEBATE ON THE TRADE AND GROWTH NEXUS *

Structure

- 16.0 Objectives
- 16.1 Introduction
- 16.2 Importance of Economic Growth
- 16.3 Sources of Economic Growth: Theoretical Underpinnings
 - 16.3.1 Theories of Growth: Classical and Endogenous Growth Models
 - 16.3.2 Trade and Growth in the Solow Model
 - 16.3.3 Trade and Productivity Growth: Theoretical Links
- 16.4 Trade Policy Regime and Growth in Developing Economies
 - 16.4.1 Criticism of Primary-Export-Led Growth Policies
 - 16.4.2 Import Substitution Industrialisation Strategy
 - 16.4.2.1 Country Experiences
 - 16.4.2.2 Criticism of ISI
 - 16.4.3 Outward Oriented Development Strategies
 - 16.4.3.1 Country Experiences
 - 16.4.3.2 Critical Evaluation
- 16.5 Indian Experience
 - 16.5.1 Experience with the Inward-Oriented Industrialisation Regime
 - 16.5.2 Experience with the Outward Oriented Liberalisation Regime
- 16.6 Let Us Sum Up
- 16.7 Key Words
- 16.8 Some Useful References
- 16.9 Answers/Hints to Check Your Progress Exercises

16.0 OBJECTIVES

After studying this Unit, you should be able to:

- Explore the relevance, nature and determinant of economic growth in general;
- Examine the relationship between international trade and economic growth in classical and endogenous growth theories;
- Identify the link between international trade and productivity growth;

* Contributed by Dr. Rijesh Raju, Institute for Studies in Industrial Development (ISID).

- Discuss the issue of trade and growth nexus in the light of trade policy regime in developing countries;
- Highlight the transition and country-level experiences of import substitution and outward-oriented policy regimes; and
- Provide an overview of the trade policy regime and growth experience of Indian economy since economic planning.

16.1 INTRODUCTION

Economic growth is of fundamental significance for any economy. An economy is said to grow when the amount of goods and services produced in the country expands over time. The total amount of output produced for final consumption in a particular period, normally a year or the economy's income level is referred to as Gross Domestic Product (GDP). Since every economy experiences growth in labour force/population, improving the rate of growth of GDP matters most for achieving sustained higher standards of living in the long run. In this sense, GDP is measured by the rate of change of real income per capita or simply, output per labour input (i.e., labour productivity). In this unit, we will learn how growth rate of an economy is related with its trade volume. We will try to understand how it affects each other.

16.2 IMPORTANCE OF ECONOMIC GROWTH

Economists argue that growth in GDP enables an economy to improve the overall welfare of the people and economic development. We know that the welfare of the citizen ultimately depends on improvement in the quality of life, including higher levels of consumption, better educational achievements, quality housing and better access to valuable health facilities. Since a sustained level of economic growth is highly correlated with higher standard of living and economic welfare, GDP growth is often considered as a key instrument (or means) for achieving economic development (the end goal).

Rather than the level of GDP, it is the rate of increase in GDP which matters most. This is because of the well-established evidence that points out that prolonged differences in GDP growth rates result in dramatic changes in living standards. One useful summary measure to appreciate the significance of rate of GDP growth is to find out the number of years it will take the size of the economy to double over time. According to a simple approximation of time, the number of years it takes for an economy to double its level is equal to 70 divided by the growth rate. This is known as "*Rule of 70*". According to this rule, an economy with a GDP growth rate of 1 per cent doubles its living standard (i.e., doubles the economy's overall size) every seventy years, while an economy with a rate of growth of 5 per cent per annum doubles its size and living standard every fourteen years.

Examples of such drastic differences in growth rates among countries are numerous. For example, in the mid-1950s, the per capita GDP of El Salvador and Japan were roughly similar. However, by 1993, the income of one Japanese person was approximately equal to that of 24 Salvadorans, which was achieved through a sustained level of per capita growth rate of less than 9 per cent per annum maintained over 38 years by the Japanese economy. This example illustrates that a percentage point drop in the growth rate makes a huge difference in the long period. Therefore, it is important to appreciate and understand the factors determining the rate of growth in GDP in an economy. In the following Section, we will attempt to provide an answer based on the standard theories of economic growth.

16.3 SOURCES OF ECONOMIC GROWTH: THEORETICAL UNDERPINNINGS

16.3.1 Theories of Growth: Classical and Endogenous Growth Models

Identifying the sources of income generation is one of the main focuses of the growth and development theories. An early attempt to understand the principal source of economic growth is attributed to the work of Prof. Robert Solow¹ in the early 1950s. The *Classical Growth Theory*, based on the work of Solow and many others, have tried to disentangle the contribution of growth in physical capital stock, growth in the labour supply and state of technology in generating income per capita over time.

Using the standard production function approach, the classical theories argue that in the long period, the rate of savings determines the size of capital stock and thus the level of output. The model predicts that an increase in saving rates will have a corresponding *level effect* on income per person (labour productivity) as output expands, but will eventually reach a steady state² equilibrium level. Alternatively, an increase in labour supply or population growth will negatively impact the steady-state levels of both capital per worker and output per worker. The Solow model predicts that regardless of the availability of factor inputs, the economy will tend to move towards a steady state long run equilibrium level. Since the level of savings determines the stock of capital, a large supply of savings allows the accumulation of productive capital and a higher output level in the steady state. However, once the economy reaches the steady state, policy

¹The Solow growth model is named after Robert M. Solow on his work on the accounting the sources of economic growth in the 1950s. The model was introduced in the paper Solow, R. M. (1956). A Contribution to the Theory of Economic Growth. *The Quarterly Journal of Economics*, 70(1), 65-94. In 1987, Solow won the Nobel Prize in economics for his work on economic growth.

² Steady state is a condition in which the key economic variables are constant and does not deviate.

instruments are needed to alter the steady-state growth rate of income per person to have a *growth-effect* over time. In effect, the policy option is to choose the steady state that provides maximum welfare to the society in terms of the availability of goods and services for consumption. The steady-state level of capital stock that maximises the level of consumption is called *Golden Rule Level of Capital*. The classical models show that if the level of capital stock is larger than the Golden rule steady state, then policies that reduce saving will increase consumption at all time periods.

However, since the capital stock is subject to decreasing returns, the capital accumulation itself cannot sustain the growth rate, as a continuous rise in output per worker requires a far greater increase in capital per worker. Naturally, over time, the economy may be subject to saving rate constraint and limit further accumulation of productive capital. In this case, for sustainable economic growth, it is necessary to have a continuous progress in the state of art of technology in production. Accordingly, the classical theories build models including exogenous technological progress, which over time expands the society's production capabilities.

A fundamental way to incorporate exogenous technological change is to assume that technology is embodied in workers skill that increases the efficiency of labour. As society's knowledge pool increases, the efficiency of labour increases and the labour per hour output of goods and services expands considerably. The exogenous rise in labour efficiency can result from new assembly line production methods, labour training, health, education skills, etc. In this *labour augmenting technological progress* model, the technological change causes the effective number of productive labours to increase over time. Therefore, at the steady state level of capital per effective labour, the rates of actual output per labour increases at the rate of labour augmenting technical progress. Thus, according to the Solow model, once the economy reaches the steady state equilibrium, the rate of growth of output per worker depends only on the rate of technological progress. In the long run, *the level of technological progress* explains sustained growth and persistent rise of standard of living in an economy.

The classical growth theories show clearly that the sustained output growth per worker requires continuous progress in the state of art of technology. The Solow model, however, considered technical change as exogenous; the model does not explain the determinant of technology. As such, the model often failed to account for many empirical realities such as cross-country income convergence, international migration pattern and large flow of capital across countries. The limitation of the Solow model in explaining the growth experiences of both the developed and the developing world prompted economists to formulate models that treat as endogenous the generation and diffusion of technology. The models that reject the over simplifying assumption of exogenous technology and explain

the determinant of technological advances is known as the *Endogenous Growth Theory*.

The endogenous theoretical models were developed since the 1980s by the scholarly writings of Paul M. Romer, Robert E. Lucas, Jr, among others.³ Instead of assuming diminishing returns for physical capital stock, the basic endogenous model incorporates knowledge as a form of capital with constant returns to scale. These models try to explain the decisions that determine knowledge creation through Research and Development (R&D). A main feature of the endogenous growth model is that they treat research as an economic activity that requires productive input resources that respond to profit opportunities. In the earlier models of Romer and others, technological progress arises from purposeful investment by the profit-seeking firms in R&D and externalities (spill over effects) from such activities. Since technological knowledge is a non-rival and partially excludable commodity, the spill over channel plays an important role in determining the rate of growth of economies over time.

16.3.2 Trade and Growth in the Solow Model

We can use the Solow model to depict the effect of international trade on economic growth. Here we will follow the model proposed by Baldwin (1992)⁴. Under the standard models of international trade, a shift from autarky to free trade increases the real income and welfare in an economy. Assuming the country produces two commodities, the Heckscher-Ohlin (H-O) model of trade shows that opening up of trade increases the real income. This is depicted by the production Possibility Frontier (PPF) movement where the nation specialises and reallocates resources in commodities that have a relative (comparative) cost advantage, determined by factor endowments. The specialisation pattern also allows consumer welfare to rise because compared to the situation in autarky, the consumption points of both commodities in free trade lie under a higher indifference curve.

According to Baldwin (1992), a one-time improvement in welfare from autarky to free trade generates a secondary improvement in real output. The secondary improvement results from increased productive efficiency that transforms available (given) inputs into welfare-enhancing final products. This productivity improvement shifts the PPF in the Solow model and generates economic growth as economy shifts up to a higher steady state equilibrium level of capital and output.

³ See Romer, P. M. (1986). Increasing Returns and Long-Run Growth. *Journal of Political Economy*, 94(5), 1002-1037; and Lucas, R. E. (1988). On the Mechanics of Economic Development. *Journal of Monetary Economics*, 22(1), 3-42.

⁴The full reference for the paper is: Baldwin, R. E. (1992). Measurable Dynamic Gains from Trade,. *Journal of Political Economy*, 100(1), 162-174.

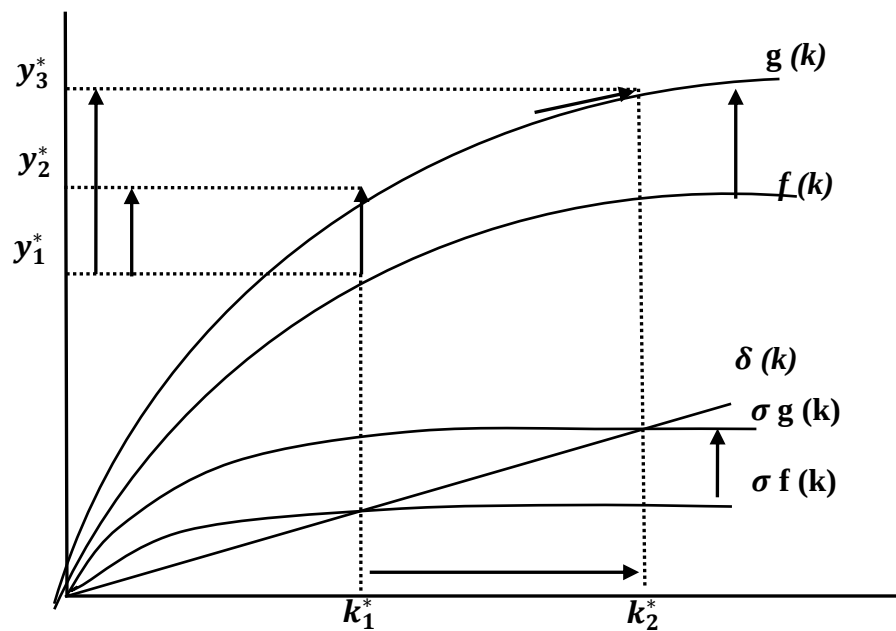


Figure 16.1 International Trade and Solow Growth Model

Source: Adapted from Berg and Lewer (2007)

The process of trade under the Solow growth model is represented in Figure 16.1. The international exchange of goods induces the PPF curve to shift from $f(k)$ to $g(k)$. This changes the economy's steady state capital-output ratio and per capita income from k_1^* and y_1^* to k_2^* and y_2^* , respectively. These changes are the result of an immediate increase in per worker output from y_1^* to y_2^* before the changes in capital-output ratio. The change in y_1^* to y_2^* is the *Static Gains from participating in International Trade*. However, the economy may further increase the gains from trade provided an increase in real income is associated with a rise in economy's saving functions. In this case, the steady state equilibrium shifts upward and generates a secondary spurt of medium-run growth that allows the economy to move to the new steady state equilibrium at k_2^* and y_3^* . The medium-run spurt may be influenced by the composition of trade, namely, whether a country's comparative advantage belongs to the production of capital goods or consumer goods. However, this model shows that the traditional gains from trade is not permanent as the economy eventually reaches and settles at the new steady state equilibrium.

However, instead of the *level effects*, the new growth theory and theoretical advancements in trade provides several plausible channels through which international trade can induce sustained *growth effects*. Based on the theoretical literature, we provide a brief overview of the plausible channels through which trade impacts productivity in the following sub-section.

16.3.3 Trade and Productivity Growth: Theoretical Links

The theoretical literature has identified several channels through which international trade impacts domestic productivity growth⁵. International trade allows domestic producers to achieve *economies of scale effects* relative to autarky. Economies of scale are realised as production moves at a lower average unit cost curve through the reduction of average fixed cost per unit of output and a downward shift of cost curve resulting from additional fixed investments made to meet the additional expectation of market opportunities abroad. Compared to autarky, a liberal trade regime induces *market competition effects* that boost firm efficiency. The entry of foreign competitive products acts as a market disciplining effect on import-competing domestic firms. For market survival, firms will be forced to invest in cost reducing activities such as technology upgradation, skill development and reduce managerial slackness. The exit of inefficient firm and reduction in monopoly power will eliminate X-inefficiency. The lower priced imports improve productivity through investment in R&D, corporate restructuring, and learning from foreign products.

Trade allows countries to specialise and follow their true comparative advantage. The traditional trade theory envisages inter-industry specialisation, whereas recent theories predict large scale within sector production reallocation from less efficient and more productive firms. The *resource reallocation effects*, induced by plant-level heterogeneity, will provide substantial dynamic productivity benefits. Since exporting involves heavy sunk-cost, it is argued that only productive firms will enter in the export market. As exporting firms share increases, the average industry productivity improves. Finally, international trade acts as a vehicle of transmission of knowledge to the recipient economy. As we noted in the earlier section, trade provides *technology spill over effects* that augment aggregate knowledge stock.

So far, we have discussed the theoretical channels through which international trade augments economic growth in an economy. In practice, countries use different policy instruments to either participate or restrict the flow of goods and services across borders. As such, the nature of trade policy initiatives adopted by the government has important implications on the pattern of trade and growth for the development of catch-up developing economies.

⁵For a detailed discussion and literature survey, see the following articles (a) Rijesh, R. (2017) International Trade and Productivity Growth: Evidence from the Organised Manufacturing Sector in India', ISID Working Paper 198, Institute for Studies in Industrial Development, New Delhi; and (b) Singh, T. (2010). Does International Trade Cause Economic Growth? A Survey. *The World Economy*, 33(11), 1517-1564.

Check Your Progress 1

Note: i) Use the space given below for your answers.

ii) Check your progress with those answers given at the end of the unit.

- 1) Why is it important to understand the factors determining the rate of growth in GDP in an economy?

.....

.....

.....

- 2) How can we use the Solow model to depict the effect of international trade on economic growth?

.....

.....

.....

- 3) Highlight the theoretical links between trade and productivity growth.

.....

.....

.....

16.4 TRADE POLICY REGIME AND GROWTH IN DEVELOPING ECONOMIES

The successful industrialisation of the western economies has forced the policy makers in the less developed economies to plan appropriate strategy for catch-up and development. Since, by definition, the less income countries have low purchasing power, the industrial development process is naturally constrained by insufficient market demand. The market constraints will lead to underutilisation of economic resources, especially in which the country has comparative advantage, especially from the availability of fertile land and resources. We know that most of the low-income countries have abundant natural resources that are ideal for producing resource-or labour-intensive products. Since increasing market access allows countries to specialise according to comparative advantage, producing and exporting labour-intensive products involves a **Primary-Export-Led Development Strategy**. This involves the support of the government to design policies that favour exploitation of natural comparative advantage in exchange of manufactured goods produced in the rest of the world.

The primary export-led growth strategy based on a country's natural economic resources is perceived to improve economic welfare as the production sector specialises and resources are reallocated according to the static gains from trade.

Moreover, trade specialisation leads to extensive use of existing factor inputs that help resources to move towards the frontier of Production Possibility Curve (PPC). This is the well-known phenomena of trade serving as a vent for surplus proposition advanced by H. Myint in 1959. The expansion of primary export may entice the inflow of foreign capital, which is relatively scarce in less-income economies that foster the recipient industry and the development of other industries. This *linkage effect* is significant in export-oriented industrialisation strategy as development creates incentive for rest of the industries to expand and build over time. This can also expand the provision of physical infrastructure such as roads, railways, harbours, telecommunications, electricity, and the like. It is argued that the primary-export-led growth strategy has been instrumental in the growth process achieved by several industrialised countries such as the United States of America, Canada, and Australia.

16.4.1 Criticism of Primary-Export-Led Growth Policies

The focus of export-led growth through specialisation in primary commodities has received considerable scrutiny and criticism from economists and government leaders. Some of the major concerns are the following.

Firstly, the market growth of primary export products is relatively lower in the industrialised countries. Based on the *Engel's law*, we know that as income increases, the demand for primary products falls relative to manufactured products.⁶ Since the developed world are the major importers there is hardly a proportionate increase in the demand for primary products, which corresponds to a declining share of industrialising countries imports in the long run.

Secondly, it is argued that in the long run, the price of primary-exported commodities relative to manufactured-goods import will tend to fall and, as a result, the exporters will face a *secular deterioration* in their terms of trade⁷. The prices of primary products fall because of the sluggish demand in the industrialised countries and over supply of these products from the severe export expansion policies in the underdeveloped countries. Under these circumstances, the unfavourable terms of trade will reduce the positive gains from trade.

⁶ Economic theory asserts that as the level of income increases, consumers spend a smaller amount of their overall income on primary commodities such as foodstuffs and other raw material based goods like textiles and apparel. Instead, their spending choices shift towards more manufactured goods. Consequently, the demand for primary products declines in relation to the demand for manufactured goods.

⁷ This thesis is proposed by (a) Singer H, W. (1950). The Distribution of Trade between Investing and Borrowing Countries. *American Economic Review*, 56, 5658. and (b) Raul Prebisch (1950), *The Economic Development of Latin America and Its Principal Problems*, United Nations.

16.4.2 Import Substitution Industrialisation Strategy

On several grounds, most developing countries ignored the primary-export-led growth strategy policy and favoured the import substitution **industrialisation (ISI)** strategies. The focus of import-substitution policies is to augment rapid industrialisation through erecting heavy protective tariff and non-tariff barriers to incentivise domestic production. The idea is to provide a necessary protective environment for the domestic industry to build manufacturing capability to learn and improve technical efficiency, productivity and create its own form of development.⁸ In essence, the import substitution represents the production of goods at home which were earlier being imported. The basic idea is rooted in the concept of *infant-industry* argument of protection for selected (or targeted) industries. This is based on the belief that the government can correctly identify the potential and emerging industrialising sectors suited for local industrialisation. Government supports the indigenous production through high trade barriers on similar products, which will encourage domestic investment in local production activities. As the industry matures, the level of protection is lowered and the sector will be able to serve the domestic and export market at competitive prices. However, in practice, the trade barriers, once erected, are difficult to repeal and don't come down easily. Another important consideration for adopting the ISI strategy was the 'Balance of Payment (BoP) difficulties' of several economies during the aftermath of World War II, as saving foreign exchange seemed to be more feasible than earning foreign exchange through export promotion.

The incidence of import substitution is often measured by a change in the ratio of imports to the total availability (i.e., import plus domestic output) of a single product or category of products. When this ratio falls over time, the particular product or sector is subject to be experiencing import substitution. As the domestic production replaces imports, the economy is expected to become more independent, resilient, diversified and welfare enhancing. In practice, most LDCs initiated ISI by using foreign exchange earnings to import capital goods such as machines and tools that make consumer or intermediate goods and raw materials to manufacture inputs to produce domestically the consumer goods previously imported. Since the terms of trade of primary export products were unfavourable, the main constraint of industrial growth and development was the availability of foreign exchange. As industrialisation requires machinery and tools, the major role of import substitution was to reduce the need for foreign exchange used to buy goods that could be made at home.

Import substitution generally goes through different stages. In the *first* stage, the government provides incentive to establish and develop final consumer goods

⁸ Bruton, H. (1989). Import substitution. *Handbook of Development Economics*, Vol 2, 1601-1644.

sectors such as food, clothing, electrical appliance, textiles, and leather to use the domestic production of consumer goods to replace imported goods. This is relatively an easy stage since the large proportion of commodities produced during this period is the consumer non-durables where comparative advantage serves as the driving force. In the *second* stage, ISI moves from consumer goods to developing capital and intermediate goods production such as machine tools, petroleum processing, iron and steel, and other capital-intensive industries. This stage is generally difficult, as the capital-intensive sectors also require advanced technology, sophisticated machinery and the availability of favourable infrastructure overheads. In the *third and final stage*, the production of complex industrial goods is envisaged (e.g., chemicals, electronic equipment, tools, etc). Once these three stages progress, industry becomes mature and lays the foundation for comprehensive industrialisation.

16.4.2.1 Country Experiences

The import substitution-based industrialisation originated in the Latin American region such as Brazil, Argentina, Mexico, etc., as a reaction against the widespread problems associated with primary export-oriented policies. Although Latin American countries were successful exporters of agricultural commodities (tropical fruits, coffee, cotton and grains) and minerals (petroleum, copper and tin) until the middle of the 20th century, these export sectors were largely controlled by foreign companies. These sectors served as foreign enclaves and had limited economic linkages with the rest of the sectors creating large income inequalities. The post-war scenario and the long-run tendency of the prices of primary commodities to fall relative to the finished manufacturing goods, as predicted by the Prebisch-Singer thesis, led the formation of an alternative economic strategy for development. As the chief advisor and head of the United Nations' Economic Commission on Latin America (ECLA), economist Prebisch argued for the case of shifting away from the dependence on raw material exports to the production of manufacture goods that substitute imports, i.e., import substitution industrialisation. Soon these policy prescriptions became the dominant paradigm followed enthusiastically by the newly independent countries in the 1950s and 1960s including India, China and other East Asian economies. In the earlier stages, the industrial sectors witnessed considerable economic growth and trade expanded rapidly. However, as countries began to shift from different *stages* of import substitution, several issues and unintended consequences emerged. Some of the leading issues and problems are discussed in the following sub-section.

16.4.2.2 Criticism of ISI

In retrospect, the development strategy based on import substitution industrialisation generated substantial economic costs including resource inefficiencies and wastages. Some of the major criticism are the following: (1) Persistent trade barriers and industry protection (2) the resource misallocation by

the government as it had commanding control over production decisions (3) severe misalignment and mismanagement of foreign exchange rate policies that made exchange rate often overvalued (4) worsening of income inequalities (5) encouraged widespread rent-seeking activities.

Under the inward oriented policy, firms were usually given temporary protection to build and develop their production capabilities. However, in practice, the barriers rarely come down. Since the level of trade/industry protection has a positive link with company profits, it is the interest of the latter to pursue and demand larger protection from domestic and international competition. As a result, the trade barriers tend to grow over time, leading to considerable welfare loss to the society.

One of the main effects of ISI is the overestimate of the government official's ability to identify market failures and solutions. Since the bureaucracy and political considerations made the resource allocation decisions, it also generated lobbying, accumulation of special provisions, favours and economic inefficiencies that build up over time. The result of heavy misallocation of scarce economic resources was magnified by the huge inequality in wealth and income throughout Latin America.

A serious issue of government controlled industrial strategy was the development and persistence of overvalued exchange rates, resulting from either deliberate policy choices or chronic problem by following a fixed exchange rate regime under inflationary pressure. The exchange rates were often overvalued to make it easier for the targeted industrial segment to import necessary capital goods and raw materials. However, on the other side, it retarded export competitiveness by raising the foreign price of domestic goods, which negatively affected the agricultural and traditional export sectors. As a result, the investments were misallocated, causing productivity and income in rural areas to remain either low or stagnant.

The exchange rate misalignment often hurt domestic manufacture that produce imperfect substitutes for imports as it was less profitable to export. Since machinery and tools become artificially cheap and easily procured from abroad, the domestic production tended to be relatively more capital intensive and insufficiently labour intensive. As a result, the manufacturing sector did not create or absorb surplus labour from agriculture and allied activities. As developed regions tend to attract more investments and infrastructure facilities, there tend to be severe economic imbalances in the contribution of subsistence farmers and industrialised class in the national income. This accentuates economic inequalities.

The concentration of economic power in a few big industrial sectors in the absence of foreign and domestic competition favoured the creation of a large number of domestic monopolies. As a result, the manufacturing activities were largely internationally uncompetitive, inefficient and inward oriented (i.e.,

catered only to the domestic demand). The small size of the domestic market hardly provided an incentive for firms to invest in new machinery and equipment or have benefits from market-induced scale economies.

Another major issue associated with the ISI was generating economic waste from rent-seeking activities. Since protection from foreign competition was the norm and domestic manufactures required import licensing and subsidised credit, the system created favourable incentive environment for private parties to form pressure group and lobby the government machinery. In effect, in the absence of a strong and transparent institutions and the pervasiveness of bribe and corruptions, the system laid ground from large-scale rent-seeking activities that diverted scarce economic resources towards unproductive activities.

16.4.3 Outward Oriented Development Strategies

In contrast to the heavy focus on internal demand-induced development model of ISI, some of the developing economies adopted ***Outward Oriented Development Strategies*** that favoured dependence on external demand opportunities. Under these policies, the government actively targeted manufacturing sectors where the country has *potential* comparative advantage. For instance, if a country was well endowed with low-or semi-skilled labour, the government would encourage the development of labour-intensive manufacturing such as textiles, paper, wood, etc., in the anticipation of better export prospects.

Under the outward looking strategy, more trade openness and less market restrictions are advocated so that internal domestic prices will be in parity to world prices. The policy also keeps an undervalued exchange rate to maintain export competitiveness and maintain relatively minimal government interference on factor market that helps wages and rents reflect true scarcity. The outward oriented policies encourage domestic manufacturers to use their relatively abundant resources and sell products abroad to achieve economies of scale. Since the success of selling products requires skill and effort to face international competitiveness, this will act as a driver to raise productivity and product quality. The export earning can be used to pay for rising imports and free the economy from binding foreign exchange constraints.

Thus, broadly, the outward oriented industrialisation policies provide similar incentives for manufacturing firms to sell either in the domestic or foreign market. However, an outward oriented policy itself is not enough to produce better growth rates as it must be supplemented by a set of policies that reduce distortions, nurture high rates of capital investments, transparent and clear property rights, rules and business regulations and an efficient infrastructure in the form of ports, airports, electricity and communications.

16.4.3.1 Country Experiences

Since the 1960s and 1970s, several developing countries shifted their emphasis toward new exports of less-skilled-labour-intensive manufactured goods to the

industrialised countries. Remarkably, the shift in industrial policy that promoted exports was followed by significant growth resurgence in several economies, especially in the East-Asian Countries. This High-Performance Asian Economies (HPAE), a term coined by the World Bank, are Hong Kong, Indonesia, Japan, Malaysia, Singapore, South Korea, Taiwan and Thailand. Among them, Japan's economy switched from ISI to export promotion in the late 1950s and early 1960s. The four Tigers (Hong Kong, South Korea, Singapore and Taiwan) initiated the transition in the late 1960s and the Newly Industrialising Economies (NIE -Indonesia, Malaysia and Thailand) began in the early 1980s. Subsequently, during 1965-2000, the HPAE doubled their share of total world exports and total world manufactured exports.

By historical standards, the growth performance achieved by NIE following changes in policy orientation toward manufactures, especially relaxing import restriction and boosting export orientation, has been phenomenal. Among the NIE, the initial shift from ISI was made by Taiwan who encouraged export promotion policies and subsequently achieved the GDP growth rates of about 10 per cent per year. South Korea shifted towards outward-oriented policies by increasing export incentives and lowering import barriers in the 1960s. Since then the economy's growth rate increased to about 10 per cent per year. Similar growth experience was observed in the case of Hong Kong and Singapore who also used export boosting policies. According to the *World Development Report of 1994* by the World Bank, during 1980-1992, the average growth rate of real GDP per person for strongly outward oriented economies (Hong Kong, South Korea and Singapore) and strongly inward oriented economies (Argentina, Bangladesh, Chile, India and Turkey) were 6.2 percent and -0.4 per cent per annum, respectively. This reflects the sharp contrast of development experience of countries following completely diverse policy strategies.

16.4.3.2 Critical Evaluation

The remarkable, consistent and robust growth of over 10 percent per annum across NIE since 1960s has been subject of considerable debate and academic research. Economists generally agree on several facts about the miracle success story of East Asian countries. *First*, these economies were subject to considerable stability in their macroeconomic indicators. *Second*, the government actively encouraged a favourable distribution of income growth, in part through enhanced access to healthcare, education, and housing. Investments in social overheads developed a highly skilled and literate labour force that improved technology and export competitiveness and attracted large-scale multi-national investments. *Third*, NIE promoted trade and not just exports, as the case with the mercantilists. Exports provided necessary foreign exchange earnings and reduced external currency constraints, while imports provided access to new technology inputs and products that encouraged productivity. *Fourth*, the growth in HPAE was accompanied by falling income inequality. Though historically most East Asian

countries have low level of unequal distribution of wealth, the subsequent policy initiatives had highly visible wealth-sharing mechanisms. These included land reforms, free public education, free basic healthcare, and considerable investments in rural infrastructure such as clean water systems, transportation and communication systems.

The improvement in economic equality is associated with rapid rates of accumulation physical capital (investment) and human capital (education and skill). The large investments were the result of considerably higher levels of saving rates that prevailed in most HPAE that was the result of rapid rate of income growth and stable macroeconomic environment of low inflation. The human capital investment was facilitated by an active education policy that created a pool of highly skilled workforce capable of managing increasingly sophisticated forms of manufacturing production. In effect, the technical skill endowment enabled most manufacturing industry to shift their level of comparative advantage (dynamic gains) so that new investments were made in new product and process lines. This supported their active promotion of export-oriented industrialisation policies.

A major learning experience of NIE success story is that apparently these economies did not follow a stringent import restrictive policy. In fact, they actively encouraged the import of essential inputs such as machinery and equipment that embodied latest technology and knowledge. The adaptation of capital goods in the domestic production boosted technological capability and exports competitiveness. This was consistent with the prediction of modern theoretical trade and economic growth models showing that protectionism often hurt manufacturing exports as the lack of demand for foreign goods forces currency to appreciate. This was in sharp contrast to the experience of Latin America where low export volumes accompanied heavy restriction on imports during the 1980s.

16.5 INDIAN EXPERIENCE

In this section, we provide a brief overview of the development experience of India since the beginning of economic planning. The focus is on the discussion of trade policy regime on industrialisation and growth.

16.5.1 Experiences with the Inward-Oriented Industrialisation Regime

Soon after independence, India adopted an inward-oriented policy regime. The basic premise was self-sufficiency, minimal external dependence and commanding heights of the public sector in the allocation of economic resources. The main policy agenda was the reservation of large segment of industries under the government ownership and the requirement of government permission before setting up industrial units by private firms. As such, the government heavily

protected the domestic industry by insulating them from international competition. This was accompanied by multiple controls over investments, scale of operations, investment location and technology to be used in production.

The trade regime was highly protectionist and regulated through tariff and pervasive non-tariff barriers on imports. Most of the imports were subject to discretionary import licensing such as Open General Licensing (OGL), negative list and canalisation by the government. On the other hand, to remove the bias from overvalued exchange rates, the government introduced several export promotion policies and instruments such as broad banding, export scrips, etc. However, the distortive incentive structure severely restricted Indian products to be internationally competitive in the world market. The blanket ISI strategy, though initially successful in developing a variety of diversified industrial base in the country, has often criticised for inefficiencies and poor rates of growth. For instance, since the mid-sixties, the rate of growth of economy was virtually stagnated around 3 per cent per annum, which came to be known as the Hindu-rate of growth. Similarly, the share of trade in GDP remained considerably low for many decades.

16.5.2 Experiences with the Outward Oriented Liberalisation Regime

The cost of blanket protection becomes apparent as the economy gradually become prey to the vicious circle of low growth, inefficiency and an uncompetitive manufacturing sector. The sluggish export sector further puts pressure on foreign exchange that was essential for meeting the essential imports. Consequently, government made efforts towards gradual liberalisation of imports in the mid-1980s. The severe BOP and fiscal crisis in the late eighties forced the government to implement comprehensive liberalisation programs to gradually reduce the over dependence on the public sector and the economic inefficiencies. The economic reforms, initiated since 1991, began to dismantle central government control in industrial activities by reducing entry barriers, abolishing the Monopolies and Restrictive Trade Practices (MRTP) Act, etc. On the external sector, considerable reduction of tariff and non-tariff barriers was made across raw materials, capital goods, intermediate goods and final goods. A managed float replaced the fixed exchange rate regime and foreign investment policies were made more market friendly and transparent. All these policy initiatives made the Indian economy more outward oriented with deeper economic integration with the rest of the world.

Since economic liberalisation, the level of trade openness, measured by merchandise trade-GDP ratio, has increased substantially from 12 per cent in

1980 to 44 per cent in 2008⁹. During 1980–2013, the organised manufacturing sector witnessed a growth rate of 8 per cent per annum with an average share of 16 per cent in GDP.¹⁰ Moreover, since the abysmally poor productivity growth rate in the nineties, there is some evidence of considerable surge in productivity in the organised manufacturing sector post 2000. Incidentally, the rate of economic growth since the 1980s has been remarkable. For instance, based on the World Bank data, we find that during the 1980s and 1990s, the GDP rose to 5.6 per cent and 5 per cent per annum, respectively. During 2000–2010 periods, the per capita GDP continued to rise at a rate of 7.7 per cent per annum, making India one of the fastest emerging economies in the developing world. This indeed suggests that there is evidence of trade led growth in India since reforms. However, much empirical evidences are required for asserting whether trade is acting as an engine of growth since the onset of comprehensive trade liberalisation reforms.

Check Your Progress 2

Note: i) Use the space given below for your answers.

ii) Check your progress with those answers given at the end of the unit.

1) Write a short note on the import substitution industrialisation strategy.

.....

.....

.....

2) Briefly explain the outward oriented development strategies.

.....

.....

.....

3) Give a brief account of India's experiences with the Inward-Oriented Industrialisation Regime.

.....

.....

.....

⁹ Trade openness is measured by taking the merchandise trade (Export-Import) share in GDP at current prices. The data is collected from World Development Indicators (WDI), World Bank online database.

¹⁰ The growth rate of manufacturing sector is based on data collected from the Annual Survey of Industries (ASI), CSO. The manufacturing share is based on value added share in GDP, collected from WDI, World Bank.

16.6 LET US SUM UP

In this unit, we studied the nexus between trade and economic growth based on international trade theory, policy and country level experiences. The impact of trade on growth is an active area of research as there are many unsettled answers posed by theory as well as country level practices. We know that trade not only improves the level of income but also can have significant growth effects through the impact on technological progress. However, since the determinant of technology is complex and specific to industry and sectors, the underlying link between trades induced investments in technology intensive activities such as R&D, skill upgradation, knowledge, etc., are not easy to comprehend. However, we do have enough evidence to argue that for the welfare benefits of liberal trade to materialise it is necessary to have an efficient, consistent and robust set of institutional arrangements governing the economic actors. Unless the rules and regulatory environment are properly defined and enforced, the gains from trade in a highly integrated economy may not be optimal.

16.7 KEY WORDS

Rule of 70

According to a simple approximation of time, the number of years it takes for an economy to double its level is equal to 70 divided by the growth rate. This is known as Rule of 70.

Golden Rule Level of Capital

The steady-state level of capital stock that maximizes the level of consumption is called Golden Rule Level of Capital.

Import Substitution Industrialization (ISI) strategies

The focus of import-substitution policies is to augment rapid industrialization through erecting heavy protective tariff and non-tariff barriers to incentivize domestic production.

Infant-industry

An infant industry is a new industry, which in its early stages experiences relative difficulty or is absolutely incapable in competing with established international competitors.

Open General Licensing (OGL)

An Open General License (OGL) is a type of export license issued by a government to its domestic suppliers.

Newly Industrializing Economies (NIE)

World Bank defines Indonesia, Malaysia and Thailand as Newly Industrializing Economies.

High-Performance Asian Economies (HPAE)

A term coined by the World Bank, HP AE considers Hong Kong, Indonesia, Japan,

16.8 SOME USEFUL REFERENCES

Gerber, James (2014) '*International Economics*', Sixth Edition, Pearson Education, Inc.

Helpman, Elhanan (2004) '*The Mystery of Economic Growth*', The Beknap Press of Harvard University Press, Cambridge, Massachusetts and London, England.

Mankiw Gregory N. (2016) '*Macroeconomics*', 9th Edition, Worth Publishers, USA.

Pugel, A.Thomas (2007), '*International Economics*' 16th Edition, McGraw-Hill Education, , New York

Sodersten Bo and Geoffrey Reed (1994) '*International Economics*', Third Edition, Macmillan Press Limited.

Steven Husted and Michael Melvin (2013) '*International Economics*', Ninth Edition, Pearson Education, Inc., publishing as Addison-Wesley.

Van den Berg, H., & Lewer, J. J. (2007) '*International Trade and Economic Growth*', Me Sharpe, Inc.

World Bank. (1993). '*The East Asian Miracle: Economic Growth and Public Policy*'. New York: Oxford University Press.

16.9 ANSWERS/HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

- 1) See Section 16.2
- 2) See sub-section 16.3.2
- 3) See sub-section 16.3.3

Check Your Progress 2

- 1) See sub-section 16.4.2
- 2) See sub-section 16.4.3
- 3) See sub-section 16.5.1

UNIT 17 TRADE AND ENVIRONMENT *

Structure

- 17.0 Objective
- 17.1 Introduction
- 17.2 Trade and Environment: Linkages
 - 17.2.1 Trade and Externalities
- 17.3 Trade and Climate Change
- 17.4 Trade and Environment: Policy and Practice
- 17.5 Role of WTO to Safeguard Environment
- 17.6 Multilateral Environment Agreements and Trade
- 17.7 Let us Sum Up
- 17.8 Key Words
- 17.9 Some Useful Readings
- 17.10 Answers/Hints to Check Your Progress Exercises

17.0 OBJECTIVES

After studying this Unit, you should be able to:

- Explain how trade and environment are related to each other;
- Describe the impact of trade on the environment;
- Discuss the policies related to trade and environment;
- Explain the role of the international agencies in protecting trade and environment; and
- Describe the various agreements signed between various nations on the area of trade and environment.

17.1 INTRODUCTION

With the increase in the volume of the world population, many issues have evolved. One of them is the relationship between trade and the environment. Is trade good or bad for the environment? The answer is not that straight. There are some environmental effects when producing goods that are exported to other nations or imported from other nations. But will these effects increase or decrease with expanded trade? What is their impact on the exporting nation, the importing

* Contributed by Prof. S.N. Mukherjee, National Institute of Public Finance and Policy

nation, or the world? And whose responsibility is it to respond to environmental hazards associated with trade? How international trading community is responding to this issue? These are some of the questions this unit will try to address.

Many emerging and developing economies often consciously adopt the export-led growth strategy in their quest for economic prosperity. A number of countries over the last two decades have achieved their economic growth by responding to the high external demand for merchandise products as well as knowledge-based services or through foreign direct investment (FDI) inflows. International trade and investment are the channels through which environmental policies transmit from one country or region to another. Although international trade is expected to harmonize environmental policies across countries in the long run, the growth process in a country may trigger sustainability challenges through various channels. The sustainability challenges may originate in various manners, namely - concentration of environmentally damaging industries (e.g., ship-breaking, toxic chemical processing), overexploitation of natural resources (e.g., surge in fishing, forestry, and mining activities), emission of greenhouse gases (GHGs) etc.

Linkages between international trade and environment are bidirectional. On one hand, stringent domestic environmental policies / regulations are often criticized for creating distortions in international trade patterns owing to the associated cost of compliance escalation and the consequent adverse implications on competitiveness. On the other hand, international trade and export liberalization policies may influence environmental sustainability through three different but integrated pathways, namely *scale effect* (attributed to accelerating economic growth), *composition effect* (attributed to structural changes in production / economy) and *technique effect* (attributed to technological progress). However, trade liberalization may not be necessarily detrimental for environment.¹

17.2 TRADE AND ENVIRONMENT: LINKAGES

First, in line with the predictions of the Environmental Kuznets Curve hypothesis (EKCH), the growing income level of a country can significantly influence its environmental sustainability. Given the availability of natural resources and labour cost advantage, foreign investment in the 'South' often extends the frontier of production and exportable surplus and brings in other concern areas. The shift in economic activities from agriculture (primary sector) to manufacturing (industrial activities) leads to large scale exploitation of both renewable (e.g., water, land, air, forest) and non-renewable (e.g., minerals, fossil fuels) natural resources, resulting in their large-scale depletion and / or

¹ For comprehensive of literature reviews on environmental impacts of international trade, see Jayadevappa and Chhatre (2000), Jeppesen *et al.* (1998), Thompson and Strohman (1996).

degradation, if adequate pollution abatement measures are not taken. The development experience of several developing countries and LDCs in Asia indicates that the industrialization drive, if not accompanied by environmental governance, often lead to sustainability challenges. Second, the Pollution Haven Hypothesis (PHH) deals with environmental degradation scenario originating from FDI inflows towards the pollution-intensive sectors / regions. Third, the Factor Endowment Hypothesis (FEH) argues that a country will specialize in the most abundantly available resource and accordingly lead to sustainability challenges. For instance, a relatively capital-abundant country may specialize in dirty capital-intensive industries, in presence of trade. However, the PHH and FEH work in opposite directions, as specialization patterns for rich (capital-abundant) and poor (labour-abundant) countries differs accordingly. Finally, over-reliance on a natural resource may lead to policy myopia and compromise the broad-based growth path, resulting in the Natural Resource Curse Hypothesis (NRCH) phenomenon. While the existence of NRCH in itself may not necessarily involve environmental challenges, specialization in 'brown' activities, particularly in mining or quarrying industries, may lead to serious concerns. These effects, either in isolation or cumulatively, often lead to not only local (e.g., release of chemical effluents) environmental problems but also trans-boundary (either regional or global) concerns (e.g., climate change concerns).

The nature of public goods of environmental problems results in market failure. It demands government interventions to set the property rights correctly and / or adopt economic policies to internalize the environmental costs.² However, inter-jurisdiction competitions to attract investment and industries, by providing economic incentives in addition to weak environmental rules and regulations and/ or lenient enforcement of the rules, often result in 'race to the bottom' competition. In other words, the lax national environmental standard in a country might function as an implicit subsidy to the local players (vis-à-vis foreign players from a country characterized by stringent environmental standards) in the sense that the marginal cost of abatement would remain below the marginal damage from pollution. However, if the export partners of a country enhance their demand for quality (e.g., through adoption of ISO 14001 standards), such trade linkages may then nullify the 'race to the bottom' forces. It has been argued that adoption of a strong environmental standard may be an optimal policy for governments, where the marginal cost of abatement would exceed the marginal environmental damage. Apart from indirect environment policy-led incentives,

² Non-inclusion of environmental costs in private marginal cost often results in shifting of environmental incidence (impacts / problems) to a large section of the society as they have to bear the costs in terms of adoption of environmental averting practices (e.g., water purifier, air purifier) and / or costs of morbidity and mortality due to environmental pollution. Therefore, the centralised private costs (the cost of pollution abatement) are shifted to public at large who may not get any benefits from the economic activity that generates pollution.

tax incentives-cum-subsidies are often used to attract investments which are not always conducive for sustainability scenario.

Since the basic trade theory does not consider environmental externalities associated with the production and/ or consumption of goods, following Harris and Roach (2017) a graphical approach of trade and environment is presented below to familiarize underlying basic concepts of trade with environmental externalities.

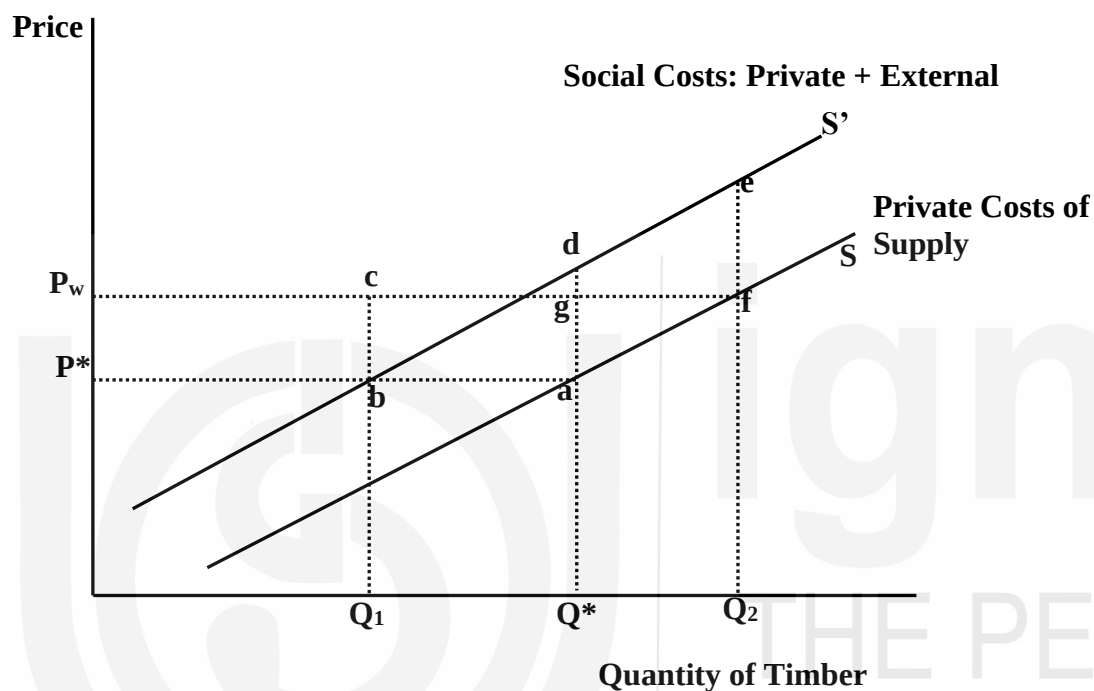


Figure 17.1 Gains and Losses from Importing Automobiles

In the absence of international trade, domestic supply and demand will be in equilibrium at 'a' and corresponding equilibrium output will be Q^* and domestic price will be P^* . If there are no barriers to trade, automobiles can be imported at the world price P_w . Domestic production will fall to Q_1 , as domestic automobile manufacturers will lose market share due to availability of cheaper imported automobiles. Domestic consumption will increase to Q_2 , as more consumers will find affordable to purchase imported automobiles. The quantity of import will be $Q_2 - Q_1$.

The domestic automobile manufacturers will lose an area P^*abP_w as they sell fewer units and that also in lower price. Domestic manufacturers will also incur loss of output equivalent to $Q^* - Q_1$. Domestic consumers will gain an area of P^*acP_w as price of automobiles fall and purchase more units. The net gain from trade will be area abc .

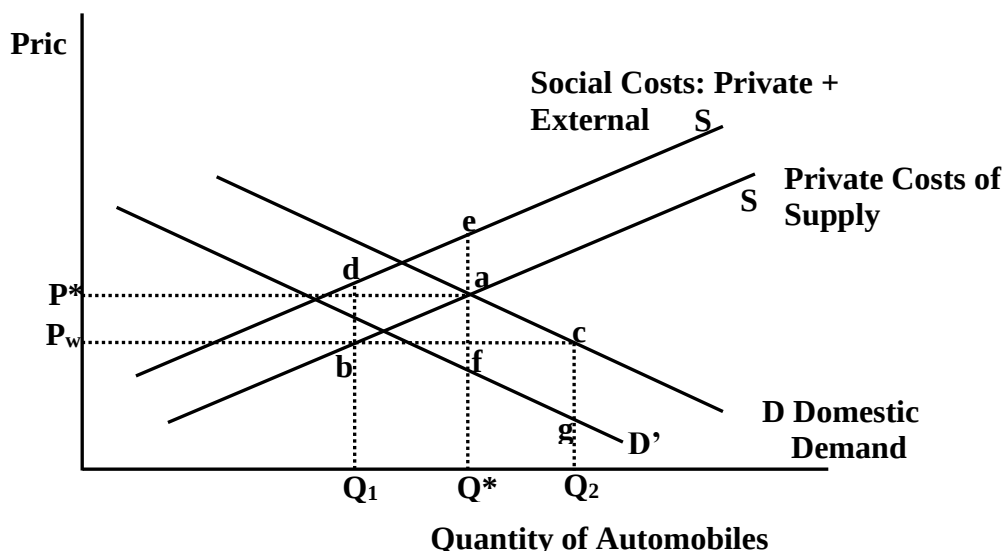


Figure 17.2: Environmental Impacts of Importing Automobiles

17.2.1 Trade and Externalities

However, this trade is not free from environmental externalities. There are two types of externalities may arise due to such trade - a) production externalities caused by production of / manufacturing of automobiles and b) consumption of externalities associated with usage of automobiles. Therefore, the gain from trade reduces if we include such externalities. Manufacturers do not include the environmental costs of production in their private cost function as a result they pass on the costs to the society in terms of air and water pollution and dumping of industrial wastes on land and water bodies. If one includes the costs associated with pollution abatement and resource recovery, costs of production will increase and the new supply curve is shown as S' in Figure 17.2. Similarly, consumers do not take into consideration the environmental externalities associated with usage of the automobile as a result it causes air pollution. If consumers are not charged for the production of externalities (exhaust emission), they pass on the costs to the society in terms of air pollution and related costs of morbidity and mortality. Therefore, benefit that accrues to consumers is reduced, as they need to incur costs to avoid exposure from pollution and/ or investing in pollution control devices (e.g., catalytic converter). Therefore, the new demand curve shifts downward at D' .

The resulting welfare impacts from trade would be area $abde$ and $afgc$ in Figure 17.2, where reduction in domestic production will result in saving in environmental costs (or environmental space) of area $abde$. In other words, the country will shift these costs to exporting country(ies) by importing cars. By importing more automobiles, the country will incur additional environmental costs of the area $afgc$ in terms of air pollution. Since, relative size of the areas ($abde$ and $afgc$) may vary depending on the environmental impacts. Therefore, it would be difficult to draw conclusions unequivocally whether trade will result in net gain or loss for a country. Moreover, it will depend on goods/commodities

involved in trade. Importing those goods whose manufacturing is detrimental for environment and there is no consumption related environmental impacts will benefit importing countries. On the other hand, importing goods whose consumption leads to environmental impacts and there is no production related environmental implication will benefit exporting countries.

To clarify consumption and production related environmental externalities and their impacts on international trade, we describe them individually below.

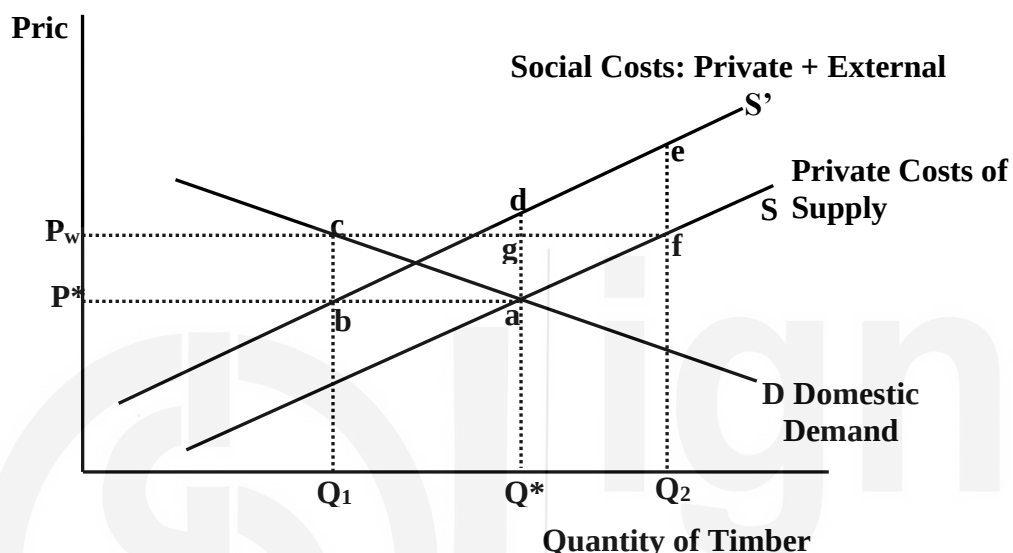


Figure 17.3: Production Externalities and International Trade

Exports of natural resources like timber and minerals result in environmental costs associated with development (extraction) of natural resources. Higher price of such natural resources in international market may lead to over-exploitation and therefore may cause large scale deforestation and rapid drawing down of mineral resources. Therefore, social costs of such natural resources are higher than private costs. Since domestic consumers have to pay higher price (P_w), they will lose consumer surplus by the area P_wcbP^* and producers of such natural resources will gain by the area P_wfaP^* . The additional gains of the producers are supported by higher price (P_w) as well as additional production of natural resources ($Q_2 - Q^*$). Additional social cost associated with such export will be the area equivalent to $adef$. There may have some social costs (externalities) associated with consumption (processing) of such natural resources in the importing countries, however for simplicity we have not considered them here. The gain from the trade - equivalent to the area $abcf$ will be weighed against additional social cost of the export - equivalent to the area $adef$. Therefore, the ultimate gain or loss from the trade will depend on relative size of the areas $abcf$ and $adef$. This example is also applicable for exports of all commodities where environmental externalities arise in the production (e.g., manufacturing textiles, leather, and metals).

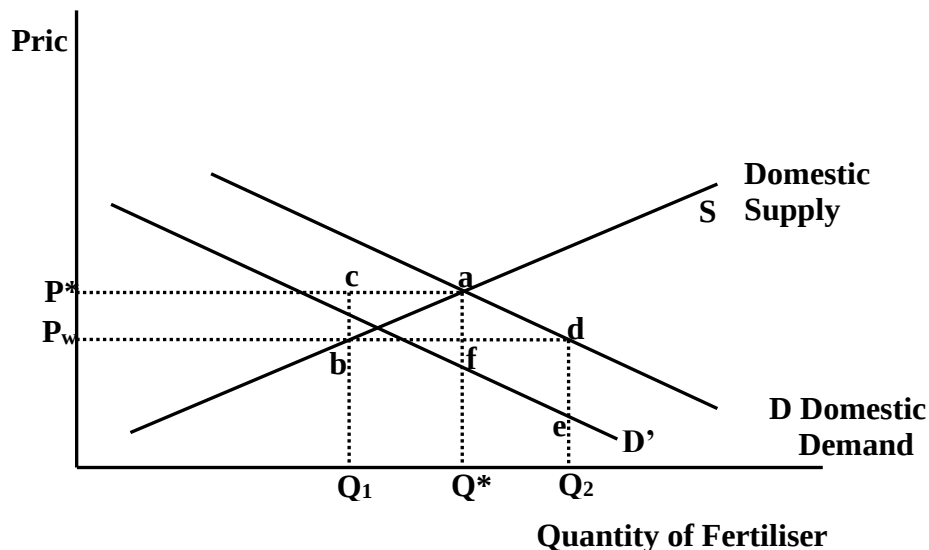


Figure 17.4: Consumption Externalities and International Trade

Import of hazardous wastes (say electronic waste) for processing and disposal is an example where severe environmental externalities are associated with demand or consumption. In the absence of international market for e-waste, domestic recyclers will pay a price P^* to procure the waste from domestic suppliers (say aggregators / waste handlers) and recycle up to Q^* (Figure 17.4). If international e-waste is available at a price of $P_w (< P^*)$, domestic recyclers will import e-waste and process up to $Q_2 (> Q^*)$. The gains of domestic recyclers will be the area adP_wP^* , where domestic aggregators will lose by the area bP_wP^*c . However, recycling of such wastes is associated with substantial environmental externalities as well as health costs to recyclers (due to exposure to heavy metals and poisonous gases). Therefore, the benefits from recycling are lower and costs associated with the externalities will be the area equivalent to $adef$. Therefore, ultimate benefits or loss associated with such type of e-waste trade will depend on relative size of the areas $adbc$ and $adef$. However, it is to be kept in mind that unless government intervenes and impose an environmental tax/ fee on such recyclers, they will succeed to retain the benefits of additional recycling of wastes and pass on the costs to the society in terms of environmental and public health impacts.

Check Your Progress 1

Note: i) Use the space given below for your answers.

ii) Check your progress with those answers given at the end of the unit.

- 1) Explain the relationship or linkages between International Trade and Environment.

.....

.....

.....

- 2) Write down a short note on Environmental Kuznets Curve hypothesis.

.....

.....

.....

- 3) Write down a short note on relationship between Consumption and production externalities with international trade.

.....

.....

.....

17.3 TRADE AND CLIMATE CHANGE

Since nineties, the globalizing forces have significantly enhanced trade flows. The growing trade has often been linked to environmental degradation (Copeland and Taylor 2004) and climate change concerns (WTO-UNEP 2009). The influence of trade flows on climate change can be explained with help of three effects. First, rise in production and exports may necessitate greater use of energy products leading to higher emissions of GHGs, which result in *scale effect*. Secondly, if a country specializes in manufacturing sectors in general and polluting sectors in particular, adverse environmental consequences may follow, leading to *composition effect*. Finally, if economic growth is accompanied by stricter emission abatement standards and stronger regulatory policies to contain the adverse climate change concerns, a more sustainable scenario might emerge through *technique effect*.

While literature on trade flows-climate change interrelationship is quite vast, the empirical evidence does not offer a clear conclusion. Cross-country studies indicate that increased outward orientation generally leads to higher GHGs emissions and environmental degradation. In addition to merchandise trade, trade in services can also significantly influence CO₂ emissions. Several country-specific studies have also analyzed the relationship between trade and climate change through time series models or case study-based approach. Evidence on emission of methane and other GHGs from agricultural and livestock exports have been noted both in developed and developing countries. Positive relationship between growing merchandise exports and increased CO₂ emissions in several developing countries has also been reported. The possible trade-emission correlation receives further support from the *Pollution Haven Hypothesis* (PHH) literature, which argues that FDI inflows in a developing country may target the pollution-intensive sectors, thereby strengthening both the *scale* and *composition effects*.

Another section of the literature however disagrees with these results. For instance, while intra-European trade flows results innovation and technology

spillovers, their trade with developing countries might lead to GHGs emissions. The PHH has been rejected by several studies arguing that it is in the best interest of the foreign investor to bring 'greener' technology, rather than extracting the advantage of lax environmental standards. Awareness is rising in many developing countries in recent period for reducing carbon emissions from trade-related activities through ISO 14000 certifications, technology upgradation etc. In all, while growing importance of *technique effect* in OECD and other high-income countries has been noted, the same might be relatively weaker for their lower income counterparts.

17.4 TRADE AND ENVIRONMENT: POLICY AND PRACTICE

The findings of the existing literature indicates that the choice of environmental policy instruments for international trade is often constrained by identification of stage of pollution generation, i.e., whether the pollution is production-based and / or consumption-induced in nature. In addition, environmental externalities of international trade could also turn out to be both production and consumption related. Trade induced *composition effects* influence pollution from production and consumption differently. Moreover, sourcing inputs for production often causes environmental problems, which are mostly local. In practice, many of the environmental impacts of international trade are local in nature (e.g., water and air pollution, land degradation, deforestation) and often exporting firms ignore local environmental problems to achieve export competitiveness. This in effect enables them to have a lower compliance cost for doing business. However, this perspective offers a moral argument to the importing countries for laying stringent certification criteria for their imports. It is argued that right choice of environmental policy instruments depends on the characteristic of the imported goods, in particular, whether they are substitutes (perfect / imperfect) or complements as well as the underlying international market condition for that product, i.e., whether perfect competition or monopoly prevails.

The implications of using environmental tax as a policy instrument have been a widely researched area. However, the advantages of linking such measures with trade have been debated. If environmental policy instruments are introduced to address the causes of environmental degradation based on the principles of environmental law (e.g., 'rectification at source', 'precautionary action', 'prevention action', 'polluter-pays'), the intervention should occur either at the production level and/ or at the consumption level rather than at the trade level. While imposition of environmental taxes improves welfare, direct setting of standards under certain circumstances may lead to welfare loss. Given the institutional capacity, input taxes on polluting inputs could achieve substantial abatement while minimizing monitoring costs. These instruments are, 'much more targeted in the sense that they are likely to achieve substantial abatement of pollution while inducing smaller welfare losses, because they have a less-

distorting impact on the rest of the economy'. The roles of structured fiscal policies (emissions taxes or their cap-and-trade equivalents with allowance auctions) for promoting greener economies are desirable. There are recommendations to impose tax on coal, oil products, and natural gas, given the potential CO₂ and local emissions from these sectors.

Import tariff reduction makes inputs cheaper, which in turn promotes production. As such a policy increases the marginal social cost; there is an incentive to impose stricter environmental regulations. Conversely, as tariff reduction lowers the revenue collection, a case for maintaining a set of weaker regulations arises that will lower the compliance cost for domestic players. While the relative strengths of the two opposing forces are ambiguous, it is expected that import tariff liberalization may not necessarily degrade the environmental scenario.

One major constraint to implement environmental policy instruments with trade objective is that they may lead to capital flight and relocation of firms, with growth implications for the host country. The perceived consequences might be more severe in case of developing countries and LDCs in general and the Asian context in particular. Evidence shows that FDI inflows have often targeted polluting industries in line with the PHH phenomenon in several South (e.g., India), East (e.g., China) and Southeast (e.g., Malaysia, Thailand, the Philippines) Asian countries. In addition, developing economies have limited institutional capacity to implement environmental policies which often constrains their ability to choose from available policy instruments for environmental protection. In face of institutional limitations, developing economies often choose tariff / trade barrier to protect domestic environment, though it cannot target pollution adequately and it has opposite effects on consumption-induced and production-based pollution.

As the converse of the tax policies, there exist a rich literature on provisions of subsidies and their implications on exports. Provision of subsidies *per se* does not imply a challenge to environmental sustainability. However, countries often provide subsidies in the area of agricultural products and natural resources, as well as offering energy subsidies to manufacturing products. Both the routes may exert an adverse impact on local sustainability as well as global climate change concerns. For instance, subsidy-led adverse environmental impacts are often noted in resource-intensive sectors like primary activities (e.g., agriculture, fisheries), transport, energy and water, etc. Also, greater fiscal devolution is often associated with rise in increasing forest land clearing for agricultural production. Fuel subsidies received by fishing trawlers and other vessels often lead to overuse of mineral fuels and over-fishing. , The provision of fuel subsidies to the iron and steel industries in China, is another case in point. On the whole, it is observed that higher budgetary spending generally leads to rise in CO₂ emissions.

In the existing literature there is no common consensus that international trade is detrimental for environment and the results are sensitive to different countries

and pollutants. For instance, there are many studies which support PHH but many studies refute it too. Often the findings are contingent on choice of methodology for data analysis. Larger data availability for a broad set of pollutants for many years and countries could facilitate generalization of the findings. In presence of these constraints, adopting proactive actions to mitigate global environmental problems like GHGs emissions often lead to adoption of environmental policy instruments to negate the environmental impacts of international trade. Underlying economic rationality behind adoption of these environmental policy instruments is that they could take into account the differences between social and private marginal costs of production relating to international trade-related activities and at the best could capture the marginal cost of abatement of pollution. However, the design of environmental policy instrument plays an important role, given its potential to distort the market conditions and possible emergence as trade barriers.

The compliance cost related concerns on competitiveness may overshadow the actions of key players, thereby affecting the process of both trade as well as climate change negotiations.

The perceived need for imposing border trade adjustments (BTAs) on imports for tackling global climate change concerns are increasingly being explored by the countries in this context and a rich literature on this front has come up. BTAs can be defined as, 'requiring trade taxes to partially reflect emission leakage effect arising from asymmetric environmental policies or, more narrowly, as requiring trade taxes to specifically embody a nation's carbon tax on the carbon content of imports'. BTAs are instruments to provide a level playing field to domestic producers where environmental standards are stringent. In practice, BTAs are charged on imports equivalent to what domestic producers pay to protect the environment and exporters are rebated for the cost they incur.

The impacts of BTAs are ambiguous as on one hand, they can potentially solve the problem of carbon leakage, encourage developing countries to protect global environment, help to achieve economic efficiency in carbon abatement, improve global welfare as it transfers carbon pricing, address the competitiveness concerns of developed countries, but also may inflict damage upon the trade of developing countries on the other. Many scholars acknowledge the importance of intervention on the part of the importing country given the fact that carbon prices differ internationally and carbon taxes in some countries might be constrained. It further argued adding to the policy debate that the, 'efficiency case for BTA depends critically on whether climate policies are pursued by carbon taxation or by cap-and-trade'. Scholars also support the imposition of BTAs, given the potential incentive for countries to weaken their standards to enhance their market access.

Check Your Progress 2

Note: i) Use the space given below for your answers.

ii) Check your progress with those answers given at the end of the unit.

- 1) Explain the role of an effective environment policy in reducing environmental hazards.

.....

.....

.....

- 2) Explain the major constraint to implement environmental policy instruments.

.....

.....

.....

17.5 ROLE OF WTO TO SAFEGUARD ENVIRONMENT

The role of WTO in achieving sustainable development and protecting environment is explained in the following WTO statement:

“Allowing for the optimal use of the world’s resources in accordance with the objective of sustainable development and seeking to protect and preserve the environment are fundamental to the WTO. These goals, enshrined in the Preamble of the Marrakesh Agreement, go hand in hand with the WTO’s objective to reduce trade barriers and eliminate discriminatory treatment in international trade relations. For WTO members, the aims of upholding and safeguarding an open and non-discriminatory multilateral trading system, on the one hand, and acting for the protection of the environment and the promotion of sustainable development, on the other, can and must be mutually supportive.”³

Ensuring that trade contributes to sustainable development is an explicit objective of the WTO and the 2030 Agenda for Sustainable Development (including the Sustainable Development Goals or ‘SDGs’). However, one area of concern is that the legal provisions of the multilateral trade regime, i.e., World Trade Organization (WTO), were not originally created to deliberate on climate change policies and sustainability challenges. While on one hand the text of the WTO Sanitary and Phytosanitary (SPS) and Technical Barriers to Trade (TBT) agreements advocate against the use of unilateral trade policies and base the

³ An introduction to trade and environment in the WTO, available online at: https://www.wto.org/english/tratop_e/envir_e/envt_intro_e.htm (last accessed on 31 July 2017).

actions on 'scientific evidence', the 'precautionary principles' based on the occurrence of 'irreversible damage' are also enshrined among the provisions.⁴ The non-recognition of 'polluter-pays-principle' by international trade institutions pose a challenge for imposing unilateral policies addressing cross-border sustainability challenges. The analysis further noted that as such bodies do not recognize the economic equivalence of emission tax and permit regimes, the resulting complexities make certain environmental policies susceptible to influence by protectionist motives. Hence the WTO-compatibility of BTAs is still open to debate.

So far, the analysis presents a case for importing countries to impose stricter environmental standards to support the climate change concerns and to protect the competitive concerns of the domestic players on the other. Nevertheless, the possibility of such measures functioning as *de facto* non-tariff barriers (NTBs) on exports of developing countries has been raised at times. In order to comply with strict environmental standards in the 'North', the producers in 'South' need to undertake compliance costs on multiple fronts, which may lead to loss of their competitiveness. If a developing country is forced to adopt a stricter environmental standard set by an external entity for exporting, its export sector suffers adversely. The underlying motive behind imposition of such standards has also been questioned. Higher standards imposed by developed countries always increase their domestic firms' profitability, but reduce the same for firms located in developing countries. In addition, it has been argued that unilateral environmental actions / sanctions are generally ineffective as an enforcement device.

The other branch of literature has noted that the stringent environmental standards have significantly improved the sustainability scenario in the South. For instance, ISO 14001 certification on environmental management awards has witnessed a 1,500 per cent increase between 1999 and 2009. The adverse effects on exports have also been questioned. It has been shown in the Chinese context that barring the exception of chemical industry, stricter environmental regulation has not adversely influenced the export potential. On the contrary, the quality of exports from both resource-intensive primary and high-value-added manufacturing sectors has improved.

In the past, trade and environment related disputes have been raised at the WTO dispute settlement forums and a 'North' move towards BTAs would only increase the number further. Given the absence of a clear understandings and agreement at the global level on managing the climate change effects through trade instruments, it is clear that the discord over environmental issues may lead

⁴ In principle, sanitary and phytosanitary (SPS) measures aim to protect the health of humans, plants and animals, while technical barriers to trade (TBT) ensure product quality and safety. However, governments may overshoot the requirements of health and consumer safety and use SPS and TBT to shield domestic producers from fair competition.

to welfare loss. The WTO case law has provided a mixed view so far. For instance, certain aspects of the verdict in DS 58⁵ pave the way for increasing incidence of unilateral trade sanctions in the future. On the other hand, by disapproving the unilateral action of the EU in the biotech case (DS 291-293)⁶, a strong message against adoption of such measures has been sent. Taking note of the legal issues yet to be resolved under WTO, the need to enable and encourage good policymaking is warranted.

The potential for a country to use voluntary export restraints (VERs) on its exports with sustainability undercurrent has recently been observed at WTO in the *rare earth* dispute. China's decision to limit export of rare earth materials, which are crucial sub-components for electronics and other high-tech products, citing their overexploitation was met by protests from countries where the domestic industries were crucially dependent on Chinese imports. Disputes were lodged at the WTO against the Chinese adoption of various policy instruments to further this goal: export duties, export quotas, export licensing and minimum export price requirements. WTO Dispute Settlement Panel ruled against China by noting that some of their instruments were geared to achieve industrial policy objectives rather than sustainability.⁷ The precedence of this case will have profound implications for other trade-environment disputes lodged at the WTO in days to come.

Protection of environment and promotion of sustainable development (SD) is one of the mandates of WTO (WTO undated). The environmental provisions are also enshrined under several WTO agreements. A change in the existing provisions is however the need of the hour. There are four specific areas for this purpose – allowing production subsidies for certain green products and technologies; arriving at a possible compromise on BTA based on carbon content in domestic production, removing / relaxing existing restrictions on exports of natural gas to facilitate substitution of dirtier substitutes like oil and coal, and tightening compulsory license mechanism for green technologies in emerging economies. The third point becomes particularly important, as a number of Asian countries

⁵ See, Dispute Settlement Case No. 58, *United States — Import Prohibition of Certain Shrimp and Shrimp Products (complaint by India; Malaysia; Pakistan; Thailand)* for details.

⁶ See, Dispute Settlement Case No. 291-293, *European Communities — Measures Affecting the Approval and Marketing of Biotech Products (complaint by United States; Canada; Argentina)* for details.

⁷ For details see, Dispute Settlement Case No. 431, *China — Measures related to the Exportation of Rare Earths, Tungsten and Molybdenum* (Complainant: United States); Dispute Settlement Case No. 432, *China — Measures related to the Exportation of Rare Earths, Tungsten and Molybdenum* (Complainant: European Union); Dispute Settlement Case No. 433, *China — Measures related to the Exportation of Rare Earths, Tungsten and Molybdenum* (Complainant: Japan).

have recently been at the WTO dispute settlement forums on the ground of support to the renewable energy sector.⁸

17.6 MULTILATERAL ENVIRONMENT AGREEMENTS AND TRADE

The role of international cooperation (agreement) is inevitable for many environmental problems. The first international treaty on trade and environment was the Phylloxera Agreement of 1978 to restrict trade in grapevines to prevent the spread of pests that damage vineyards. In response to emerging environmental challenges, a large number of multilateral environmental agreements (MEAs) were mooted to minimize the trade related impacts on environment and natural resources. These agreements played important role in conservation of natural resources and biodiversity, protecting endangered species, and minimizing spillover of hazardous substances and environmental problems across countries. About 17 MEAs were identified by GATT (1992) to contain trade measures, starting in 1933. However, each of these have not ratified by all the countries. Major MEAs having bearing on trade and environment are as follows:

- International Convention for the Protection of Birds (1950)
- International Plant Protection Agreement (1951)
- Convention on Conservation of North Pacific Fur Seals (1957)
- Agreement Concerning Cooperation in the Quarantine of Plants and their Protection against Pests and Diseases (1959)
- Rio International Convention for the Conservation of Atlantic Tunas (ICCAT) (1966)
- Phyto-sanitary Convention for Africa (1967)
- The African Convention on the Conservation of Nature and Natural Resources (1968)
- The Benelux Convention of on the Hunting and Protection of Birds (1970)
- Convention on International Trade in Endangered Species (CITES) of Wild Fauna and Flora (1973)
- Montreal Protocol on Substances that Deplete the Ozone Layer (1987)

⁸ For details see, Dispute Settlement Case No. 412, *Canada — Certain Measures Affecting the Renewable Energy Generation Sector (complaint by Japan)*; Dispute Settlement Case No. 419, *China — Measures concerning wind power equipment (complaint by United States)*; Dispute Settlement Case No. 456, *India — Certain Measures Relating to Solar Cells and Solar Modules (Complainant: United States)*.

- Basel Convention on the Control of Transboundary Movements of Hazardous Wastes and their Disposal (1989)
- Convention for the Prohibition of Fishing with Long Driftness in the South Pacific (Wellington Convention) (1990)
- Antarctica Treaty (1991)
- Convention on Biological Diversity (1992)
- Convention on Straddling and Highly Migratory Fish Stocks (1995)
- Kyoto Protocol on Climate Change (1997)
- Stockholm Convention on Persistent Organic Pollutants (2001)
- Convention on Biological Diversity (2002)
- Cartagena Protocol on Biosafety (2003)

Since all MEAs are not compatible with WTO rules, the precedence of international agreements in case of conflict is a matter of debate.

Check Your Progress 3

Note: i) Use the space given below for your answers.

ii) Check your progress with those answers given at the end of the unit.

1) Comment on the role of WTO in safeguarding environment..

.....

.....

.....

2) Explain the major multilateral agreements to control environmental degradation.

.....

.....

.....

17.7 LET US SUM UP

In this unit, we discussed about linkages between trade and environment. International trade itself may not automatically correct the distortions in prices, thereby making environmental policy instruments one of the options available. Achieving high economic growth and exports, creating jobs, etc. are often the foremost priorities compared to protecting the environment for many developing and emerging economies in the world, and the Asian economies are no exception to this trend. Therefore, there is lack of seriousness to implement commitments made in multilateral environmental agreements. Though regional forums exist to discuss sustainability concerns, the working of the same in the South is still in the

evolving stage. While environmental sustainability has been included among the areas of cooperation in the Asian regional trade agreements (RTAs), they are still not appropriate forums to hold a country for particular domestic environmental concerns. The contrast with inclusion of environmental provisions under North American Free Trade Agreement (NAFTA) or the EU Economic Partnership Agreements (EPAs) is clearly observed. This is one area for the Asian countries to deepen regional cooperation. Regional Comprehensive Economic Partnership (RCEP) may play a key role in this regard.

As several Asian countries are now facing environmental compliance requirements in developed countries, which often function as NTBs their exports, there is a visible move towards carbon tax regimes - e.g., pilot emissions trading schemes (ETS) at municipal and provincial levels in China, introduction of ETS in South Korea, introduction of the tax for climate change mitigation and voluntary emissions trading scheme in Japan, implementation of clean development mechanism (CDM) projects and a domestic voluntary carbon market in Vietnam, introduction of coal cess and pilot ETS for air pollutants in India, the proposed ETS in Indonesia and Thailand. It is expected that in the coming days, more Asian countries would be interested to tread along the path of environmental policy instruments for two reasons. First, as the United Nations Framework Convention on Climate Change (UNFCCC) negotiation intensifies, the Asian countries would also be compelled to fulfill their commitments. Second, the need to secure the export market may cause them to tighten their environmental norms and policy instruments, increasingly aligning the same with the requirements prevailing in their developed counterparts.

17.8 KEY WORDS

Environmental hazards

An environmental hazard is a substance, state or event which has the potential to threaten the surrounding natural environment / or adversely affect people's health, including pollution and natural disasters such as storms and earthquakes.

Greenhouse gases

A greenhouse gas is a gas in an atmosphere that absorbs and emits radiant energy within the thermal infrared range. This process is the fundamental cause of the greenhouse effect.

Environmental Kuznets Curve hypothesis

The Environmental Kuznets Curve (EKC) hypothesis postulates an inverted-U-shaped relationship between different pollutants and per capita income, i.e., environmental pressure increases up to a certain level as income goes up; after that, it decreases

Pollution Haven Hypothesis

The pollution haven hypothesis posits that, when large industrialized nations seek to set up factories or offices abroad, they will often look for the cheapest option in terms of resources and labor that offers the land and material access they require.

Technical Barriers to Trade

A category of nontariff barriers to trade, are the widely divergent measures that countries use to regulate markets, protect their consumers, or preserve their natural resources, but they also can be used to discriminate against imports in order to protect domestic industries.

Environmental externalities

Environmental externalities refer to the economic concept of uncompensated environmental effects of production and consumption that affect consumer utility and enterprise cost outside the market mechanism. As a consequence of negative externalities, private costs of production tend to be lower than its “social” cost.

17.9 SOME USEFUL REFERENCES

Harris, Jonathan and Brian Roach (2017), ‘Environmental and Natural resources Economics: A Contemporary Approach’, *Chapter 20: World Trade and the Environment*, Routledge: London & New York.

Rao, P. K. (2000), “Sustainable Development: Economics and Policy”, Chapter 8: Debt, Trade and Environment, Oxford: Blackwell Publishers.

Copeland, B. R. and Taylor, M. S. (2004) ‘Trade, Growth, and the Environment’, *Journal of Economic Literature*, 42(1): 7-71

Harris, Jonathan and Brian Roach (2017), ‘Environmental and Natural resources Economics: A Contemporary Approach’, *Chapter 20: World Trade and the Environment*, Routledge: London & New York.

Jayadevappa, R. and Chhatre, S. (2000), "International trade and environmental quality: a survey", *Ecological Economics*, 32: 175–194.

Jeppesen, T., Folmer, H. and Komen, R. (1998) ‘Impacts of Environmental Policy on International Trade and Capital Movement: A Synopsis of the Macroeconomic Literature’, paper presented at the World Congress of Environmental and Resource Economists, 25-27th June, Venice, Italy.

O'Connor, D. (2004) 'OECD Country Economic and Environmental Policies: What Implications for East Asia?', Prepared for Experts' Seminar on The Impact and Coherence of OECD Country Policies on Asian Developing Economies, OECD, Paris.

Thompson, P. and Strohm, L. A. (1996) 'Trade and Environmental Quality: A Review of the Evidence', *Journal of Environment & Development*, 5(4): 363-388.

World Trade Organisation (undated) 'Early years: emerging environment debate in GATT/WTO', available at: https://www.wto.org/english/tratop_e/envir_e/hist1_e.htm (accessed 27 September 2015).

World Trade Organisation and United Nations Environment Programme (2009) 'Trade and Climate Change', Geneva: WTO and UNEP.

17.10 ANSWERS/HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

- 1) Refer to Section 17.2
- 2) Refer to Section 17.2

Check Your Progress 2

- 1) Refer to Section 17.4
- 2) Refer to Section 17.4

Check Your Progress 3

- 1) Refer to Section 17.5
- 2) Refer to Section 17.6